

Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

Erik Brynjolfsson* Bharat Chandar† Ruyu Chen‡§¶

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Abstract

This paper examines changes in the labor market for occupations exposed to generative artificial intelligence using high-frequency administrative data from the largest payroll software provider in the United States. We present six facts that characterize these shifts. We find that since the widespread adoption of generative AI, early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 13 percent relative decline in employment even after controlling for firm-level shocks. In contrast, employment for workers in less exposed fields and more experienced workers in the same occupations has remained stable or continued to grow. We also find that adjustments occur primarily through employment rather than compensation. Furthermore, employment declines are concentrated in occupations where AI is more likely to *automate*, rather than *augment*, human labor. Our results are robust to alternative explanations, such as excluding technology-related firms and excluding occupations amenable to remote work. These six facts provide early, large-scale evidence consistent with the hypothesis that the AI revolution is beginning to have a significant and disproportionate impact on entry-level workers in the American labor market.

*Stanford University and NBER; erikb@stanford.edu

†Stanford University; chandarb@stanford.edu

‡Stanford University; ruyuchen@stanford.edu

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1 Introduction

The proliferation of generative artificial intelligence (AI) has sparked a global debate about its potential impact on the labor market. This discourse, across academia, public policy, business, and popular media, spans utopian predictions of enhanced productivity, dystopian fears of widespread job displacement, and skeptical views that AI will have minimal effects on employment or productivity. Historically, technologies have affected different tasks, occupations, and industries in different ways, replacing work in some, augmenting others, and transforming still others. These heterogeneous effects suggest that there may be “canaries in the coal mine” which are harbingers of more widespread effects of AI.

There have been rapid improvements in AI capabilities in several areas. For instance, according to the most recent AI Index Report, AI systems could solve just 4.4% of coding problems on SWE-Bench, a widely used benchmark for software engineering, in 2023, but performance increased to 71.7% in 2024 (Maslej et al., 2025).¹ AI has improved on other benchmarks as well including language understanding, subject knowledge, and reasoning. At the same time, AI systems are increasingly widely adopted. According to Hartley et al. (2025), LLM adoption at work among U.S. survey respondents above age 18 reached 46% by June/July 2025.²

Given the better capabilities and widespread adoption, a central concern, amplified in recent headlines, is whether AI is beginning to supplant human labor, particularly for younger, entry-level workers in highly exposed professions like software engineering and customer service.³

Despite the intensity of this debate, empirical evidence has struggled to keep pace with technological advancement, leaving many fundamental questions unanswered. This paper confronts this empirical gap by leveraging a large-scale, high-frequency administrative dataset from ADP, the largest payroll software provider in the United States. Our sample consists of monthly, individual-level payroll records through July 2025, encompassing millions of workers across tens of thousands of firms. This rich panel structure allows us to track employment dynamics with a high degree of

¹SWE-bench is designed to evaluate the performance of large language models (LLMs) on real-world software engineering tasks. It uses a collection of GitHub issues to assess an LLM’s ability to generate code that resolves those issues.

²Similarly, Bick et al. (2024) found that in late 2024, nearly 40% of the U.S. population age 18-64 reported using generative AI, with 23% of employed respondents saying they had used generative AI for work at least once in the previous week, and 9% every work day.

³Improved productivity of workers in an occupation could lead to either reduced or increased employment, depending on, among other things, how elastic demand is for the output of those workers.

granularity, providing a near real-time view of labor market adjustments. By linking this data to established measures of occupational AI exposure and other variables, we can quantify the realized employment changes since the widespread adoption of generative AI.

This paper systematically presents six key facts that emerge from the data, offering an assessment of how the AI revolution is reshaping the American workforce.

Our first key finding is that we uncover substantial declines in employment for early-career workers (ages 22-25) in occupations most exposed to AI, such as software developers and customer service representatives. In contrast, employment trends for more experienced workers in the same occupations, and workers of all ages in less-exposed occupations such as nursing aides, have remained stable or continued to grow.

Our second key fact is that overall employment continues to grow robustly, but employment growth for young workers in particular has been stagnant since late 2022. In jobs less exposed to AI young workers have experienced comparable employment growth to older workers. In contrast, workers aged 22 to 25 have experienced a 6% decline in employment from late 2022 to July 2025 in the most AI-exposed occupations, compared to a 6-9% increase for older workers. These results suggest that declining employment AI-exposed jobs is driving tepid overall employment growth for 22- to 25- year-olds as employment for older workers continues to grow.

Our third key fact is that not all uses of AI are associated with declines in employment. In particular, entry-level employment has declined in applications of AI that *automate* work, but not those that most *augment* it. We distinguish between automation and augmentation empirically using estimates of the extent to which observed queries to Claude, the LLM, substitute or complement for the tasks in that occupation. While we find employment declines for young workers in occupations where AI primarily automates work, we find employment growth in occupations in which AI use is most augmentative. These findings are consistent with automative uses of AI substituting for labor while augmentative uses do not.

Fourth, we find that employment declines for young, AI-exposed workers remain after conditioning on firm-time effects. One class of explanations for our patterns is that they may be driven by industry- or firm-level shocks such as interest rate changes that correlate with sorting patterns by age and measured AI exposure. We test for a class of such confounders by controlling for firm-time effects in an event study regression, absorbing aggregate firm shocks that impact all workers at a

Quarterly Census of Employment and Wages (QCEW).¹⁰ These prior papers use data that lack either sufficient granularity or immediacy to reliably study employment changes by AI exposure and age (O’Brien, 2025).¹¹ In contrast, this paper uses large-scale, close to real-time data to take a step towards resolving the ongoing debate on the employment effects of AI on young workers.

3 Data Description

3.1 Payroll Data

This study uses data from ADP, the largest payroll processing firm in America. The company provides payroll services for firms employing over 25 million workers in the US. We use this information to track employment changes for workers in occupations measured as more or less exposed to artificial intelligence.

We make several sample restrictions for our main analysis sample. We include only workers employed by firms that use ADP’s payroll product to maintain worker earnings records. We also exclude employees classified by firms as part-time from the analysis and subset to people between the age of 18 and 70.¹²

The set of firms using payroll services changes over time as companies join or leave ADP’s platform. We maintain a consistent set of firms across our main sample period by keeping only companies that have employee earnings records for each month from January 2021 through July 2025.

In addition, ADP observes job titles for about 70% of workers in its system. We exclude workers who do not have a recorded job title. There are over 7,000 standardized job titles, examples of which include “Search engineer optimization specialist,” “Enterprise content management manager,” and “Plant documentation control specialist.” The company’s internal research team maps each of

¹⁰Johnston and Makridis (2025) measure state-industry exposure by taking an average of Eloundou et al. (2024)’s occupational exposure weighted by state-industry employment. Industry-level labor market changes may be distinct from the occupation-level changes studied in this paper if firms make capital investments or become more productive in ways that increases overall labor demand (Hampole et al., 2025).

¹¹As a comparison, the CPS surveyed between 44,000 and 51,000 employed individuals in total across all age groups in each month since 2021. Between 10,000 and 12,000 of these observations were in the outgoing rotation group and included earnings records. The data in our main analysis sample includes between 250,000 and 350,000 employed individuals in each month *just between the ages of 22 and 25*, all with earnings records.

¹²While we observe the year of birth for each worker, for privacy reasons we do not observe the exact date of birth. We impute month of birth from the distribution of birth months in the United States using data from the Center for Disease Control and Prevention.

4.5 Fact 5: Labor market adjustments are visible in employment more than compensation

In addition to employment we observe workers’ annual base compensation. We use this information to test for labor market adjustment along the compensation margin.¹⁹ Salary data are deflated to 2017 dollars using the PCE index.²⁰

Results are in Figure 10. The findings indicate a less marked divergence in compensation compared to employment across more and less exposed occupations. Figure 11 shows results by age and Eloundou et al. (2024)-based exposure quintile. We find little difference in compensation trends by age or exposure quintile.

Prior work by Autor and Thompson (2025) notes that technology that replaces inexpert tasks may reduce occupational employment but increase occupational wages; technology that replaces expert tasks may do the opposite. The sign of the wage effect depends on the overall share of tasks displaced as well as the whether these tasks are expert or inexpert. The limited changes we find for wages suggest that these effects may be offsetting, at least in the short run. Alternatively, the results could be explained by wage stickiness in the short run, consistent with recent evidence from Davis and Krolikowski (2025).

4.6 Fact 6: Findings are largely consistent under alternative sample constructions

We test the robustness of these results to alternative sample constructions and robustness checks.

Excluding Technology Occupations One possibility is that our results are explained by a general slowdown in technology hiring from 2022 to 2023 as firms recovered from the COVID-19 Pandemic.²¹ Figure A4 shows employment changes by age and exposure quintile after excluding computer occupations, corresponding to 2010 SOC codes that start with 15-1. Figure A5 shows results when excluding firms in the information sector (NAICS code 51). Results are quite similar,

¹⁹Total compensation may additionally include bonuses, overtime pay, commissions, equity, tips, and other items. These may have a greater impact on overall compensation in certain professions and age groups than others.

²⁰In contrast to the series for employment, results for compensation end in June 2025, the most recently available month for the PCE index.

²¹Under the Tax Cuts and Jobs Act, amendments to Internal Revenue Code §174 enacted in 2022 also disallowed companies from immediately deducting R&D expenditures, including software development costs. These costs instead had to be capitalized and amortized over five years for domestic research and fifteen years for foreign research.